

Darboux's formula

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1 Darboux's formula

Let $\phi(t)$ be a polynomial of degree n and $f(x)$ is an analytic function then,

Theorem 1.1.

$$\sum_{m=0}^n (-1)^m (z-a)^m \left[\phi^{(n-m)}(1) f^{(m)}(z) - \phi^{(n-m)}(0) f^{(m)}(a) \right] = (-1)^n (z-a)^{n+1} \int_0^1 \phi(t) f^{(n+1)}[a+t(z-a)] dt \quad (1)$$

Proof

For $n = 0$ we have $\phi(t) = c$ then

$$LHS_0 = c [f(z) - f(a)]$$
$$RHS_0 = (z-a) \int_0^1 c f^{(1)}[a+t(z-a)] dt$$

After a change of variables $u = a + t(z-a)$, $du = dt(z-a)$ we get

$$RHS_0 = c \int_a^z f^{(1)}(u) du = c [f(u)] \Big|_{u=a}^{u=z} = c [f(z) - f(a)] = LHS_0$$

For $n > 0$

$$RHS_n = (-1)^n (z-a)^{n+1} \int_0^1 \phi(t) f^{(n+1)}[a+t(z-a)] dt$$

Using integration by parts $\int_{t=0}^1 uv' = uv \Big|_{t=0}^1 - \int_{t=0}^1 u'v$, with $u = \phi(t)$, and $v' = f^{(n+1)}(a+t(z-a))$

$$RHS_n = (-1)^n (z-a)^{n+1} \left[\phi(t) \frac{f^{(n)}[a+t(z-a)]}{(z-a)} \Big|_{t=0}^{t=1} - \int_0^1 \phi'(t) \frac{f^{(n)}[a+t(z-a)]}{(z-a)} dt \right]$$

Evaluating the first definite integral and multiplying by $(z-a)$

$$RHS_n = (-1)^n (z-a)^n \left[\phi(1) f^{(n)}(z) - \phi(0) f^{(n)}(a) - \int_0^1 \phi'(t) f^{(n)}[a+t(z-a)] dt \right]$$

Moving the second integral out to it's own term

$$RHS_n = (-1)^n (z-a)^n \left[\phi(1)f^{(n)}(z) - \phi(0)f^{(n)}(a) \right] + (-1)^{n-1} (z-a)^n \int_0^1 \phi'(t)f^{(n)}[a+t(z-a)] dt$$

Repeating the integration by parts process we get

$$\begin{aligned} RHS_n &= (-1)^n (z-a)^n \left[\phi(1)f^{(n)}(z) - \phi(0)f^{(n)}(a) \right] \\ &+ (-1)^{n-1} (z-a)^{n-1} \left[\phi^{(1)}(1)f^{(n-1)}(z) - \phi^{(1)}(0)f^{(n-1)}(a) \right] \\ &+ (-1)^{n-2} (z-a)^{n-1} \int_0^1 \phi^{(2)}(t)f^{(n-1)}[a+t(z-a)] dt \end{aligned}$$

Repeating the integration by parts n -times until $\phi^{(n+1)}(t) = 0$ we get

$$RHS_n = \sum_{k=0}^n (-1)^{n-k} (z-a)^{n-k} \left[\phi^{(k)}(1)f^{(n-k)}(z) - \phi^{(k)}(0)f^{(n-k)}(a) \right] + 0$$

Changing the index of summation $m = n - k$

$$RHS_n = \sum_{m=0}^n (-1)^m (z-a)^m \left[\phi^{(n-m)}(1)f^{(m)}(z) - \phi^{(n-m)}(0)f^{(m)}(a) \right]$$

Therefore we get

$$\begin{aligned} \sum_{m=0}^n (-1)^m (z-a)^m \left[\phi^{(n-m)}(1)f^{(m)}(z) - \phi^{(n-m)}(0)f^{(m)}(a) \right] = \\ (-1)^n (z-a)^{n+1} \int_0^1 \phi(t)f^{(n+1)}[a+t(z-a)] dt \end{aligned}$$

2 Taylor's formula

Set $\phi(t) = (t - 1)^n$ into Darnoux's formula to obtain Taylor's formula.

Theorem 2.1.

$$f(z) = \sum_{m=0}^n \frac{(z-a)^m f^{(m)}(a)}{m!} + (-1)^n \frac{(z-a)^{n+1}}{n!} \int_0^1 (t-1)^n f^{(n+1)}[a+t(z-a)] dt \quad (2)$$

Proof

Starting from Darboux's formula ??

$$\begin{aligned} \sum_{m=0}^n (-1)^m (z-a)^m \left[\phi^{(n-m)}(1) f^{(m)}(z) - \phi^{(n-m)}(0) f^{(m)}(a) \right] = \\ (-1)^n (z-a)^{n+1} \int_0^1 \phi(t) f^{(n+1)}[a+t(z-a)] dt \end{aligned}$$

Setting $\phi(t) = (t-1)^n$, which is an n -degree polynomial

Note that $\phi^{(k)}(t) = \frac{n!}{(n-k)!} (t-1)^{n-k}$ for $k \leq n$, therefore

$$\begin{aligned} \phi^{(k)}(0) &= \frac{n!}{(n-k)!} (-1)^{n-k} \quad \text{for } k \leq n \\ \phi^{(k)}(1) &= 0 \quad \text{for } k < n \\ \phi^{(n)}(1) &= n! \end{aligned}$$

$$\begin{aligned} \sum_{m=1}^n (-1)^m (z-a)^m \left[0 - \frac{n!(-1)^m}{(m)!} f^{(m)}(a) \right] + (-1)^0 (z-a)^0 \left[n! f^{(0)}(z) - n! f^{(0)}(a) \right] \\ = (-1)^n (z-a)^{n+1} \int_0^1 (t-1)^n f^{(n+1)}[a+t(z-a)] dt \end{aligned}$$

Simplifying and dividing by $n!$

$$\begin{aligned} - \sum_{m=1}^n \frac{(z-a)^m}{m!} f^{(m)}(a) + [f(z) - f(a)] \\ = (-1)^n \frac{(z-a)^{n+1}}{n!} \int_0^1 (t-1)^n f^{(n+1)}[a+t(z-a)] dt \end{aligned}$$

Rearranging

$$f(z) = f(a) + \sum_{m=1}^n \frac{(z-a)^m}{m!} f^{(m)}(a) + (-1)^n \frac{(z-a)^{n+1}}{n!} \int_0^1 (t-1)^n f^{(n+1)}[a+t(z-a)] dt$$

Adding the extra term into the summation to finish the proof

$$f(z) = \sum_{m=0}^n \frac{(z-a)^m}{m!} f^{(m)}(a) + (-1)^n \frac{(z-a)^{n+1}}{n!} \int_0^1 (t-1)^n f^{(n+1)}[a+t(z-a)] dt$$

2.1 Infinite Series

Theorem 2.2. If $|f^{(k)}(x)| \leq M$ and $|z - a| \leq R \quad \forall x$ then

$$\boxed{f(z) = \sum_{m=0}^{\infty} \frac{(z-a)^m f^{(m)}(a)}{m!}} \quad (3)$$

Proof

Starting from Taylor's formula ?? and taking the limit as n approaches infinity

$$f(z) = \sum_{m=0}^{\infty} \frac{(z-a)^m f^{(m)}(a)}{m!} + \lim_{n \rightarrow \infty} (-1)^n \frac{(z-a)^{n+1}}{n!} \int_0^1 (t-1)^n f^{(n+1)}[a + t(z-a)] dt$$

Define the remainder term, and then we will show that it converges to 0

$$\begin{aligned} R_n &= (-1)^n \frac{(z-a)^{n+1}}{n!} \int_0^1 (t-1)^n f^{(n+1)}[a + t(z-a)] dt \\ |R_n| &= \left| (-1)^n \right| \cdot \left| \frac{(z-a)^{n+1}}{n!} \right| \cdot \left| \int_0^1 (t-1)^n f^{(n+1)}[a + t(z-a)] dt \right| \\ |R_n| &\leq \frac{|z-a|^{n+1}}{n!} \cdot \int_0^1 \left| (t-1)^n \right| \cdot \left| f^{(n+1)}[a + t(z-a)] \right| dt \end{aligned}$$

Using $|f^{(k)}(x)| \leq M$

$$|R_n| \leq \frac{|z-a|^{n+1}}{n!} \cdot \int_0^1 |t-1|^n \cdot M dt$$

Using $|z-a| \leq R$

$$|R_n| \leq \frac{R^{n+1}M}{n!} \cdot \int_0^1 |t-1|^n dt$$

Note that between $0 \leq t \leq 1$, $|t-1| \leq 1$, therefore

$$|R_n| \leq \frac{R^{n+1}M}{n!}$$

Therefore since M and R are constant:

$$\lim_{n \rightarrow \infty} R_n \leq \lim_{n \rightarrow \infty} |R_n| \leq 0$$

Therefore

$$f(z) = \sum_{m=0}^{\infty} \frac{(z-a)^m f^{(m)}(a)}{m!}$$

3 Euler-Maclaurin formula

Set $\phi(t) = B_n(x)$ into Darnoux's formula to obtain the Euler-Maclaurin formula, where $B_n(x)$ is the Bernoulli polynomial¹.

Theorem 3.1. *For any analytical $g(x)$*

$$\boxed{\begin{aligned} w \sum_{j=0}^{r-1} g(a+jw) - \int_{x=a}^{a+rw} g(x)dx &= \frac{w}{2} (g(a) - g(a+rw)) + \\ &\sum_{k=1}^{\lfloor n/2 \rfloor} \frac{w^{2k} B_{2k}}{(2k)!} (g^{(2k-1)}(a+rw) - g^{(2k-1)}(a)) + R_{n,r} \\ R_{n,r} &= \frac{(-1)^{n+1} w^{n+1}}{n!} \int_0^r B_n(\{t\}) f^{(n+1)}(a+tw) dt \end{aligned}} \quad (4)$$

Proof

Starting from Darboux's formula ??

$$\begin{aligned} \sum_{m=0}^n (-1)^m (z-a)^m \left[\phi^{(n-m)}(1) f^{(m)}(z) - \phi^{(n-m)}(0) f^{(m)}(a) \right] = \\ (-1)^n (z-a)^{n+1} \int_0^1 \phi(t) f^{(n+1)}[a+t(z-a)] dt \end{aligned}$$

Setting $\phi(t) = B_n(t)$, which is an n -degree polynomial

Note that $\phi^{(k)}(t) = \frac{n!}{(n-k)!} B_{n-k}(t)$ for $k \leq n$, therefore

$$\begin{aligned} \phi^{(k)}(0) = \phi^{(k)}(1) &= \frac{n!}{(n-k)!} B_{n-k} \quad \text{for } k \leq n, k \neq 1 \\ \phi^{(1)}(0) = -\phi^{(1)}(1) &= B_1 \end{aligned}$$

adjusting the indicies we get

$$\begin{aligned} \phi^{(n-m)}(0) = \phi^{(n-m)}(1) &= \frac{n!}{m!} B_m \quad \text{for } n \geq m \geq 0, m \neq 1 \\ \phi^{(n-1)}(0) = B_1 &= -\phi^{(n-1)}(1) \end{aligned}$$

Plugging these values into Darboux's formula, and removing the first two terms we get

$$\begin{aligned} \sum_{m=2}^n (-1)^m (z-a)^m \left[\frac{n!}{m!} B_m f^{(m)}(z) - \frac{n!}{m!} B_m f^{(m)}(a) \right] + n! B_0 (f(z) - f(a)) \\ + (-1)^n n! (z-a) \left(-B_1 f'(z) - B_1 f'(a) \right) = (-1)^n (z-a)^{n+1} \int_0^1 B_n(t) f^{(n+1)}[a+t(z-a)] dt \end{aligned}$$

¹ $B_n(x) = \sum_{k=0}^n \binom{n}{k} B_k x^{n-k}$. See the Bernoulli number document for the full definition.

simplifying

$$\sum_{m=2}^n \frac{(-1)^m (z-a)^m B_m}{m!} (f^{(m)}(z) - f^{(m)}(a)) + B_0 (f(z) - f(a)) + (z-a) B_1 (f'(z) + f'(a)) = \frac{(-1)^n (z-a)^{n+1}}{n!} \int_0^1 B_n(t) f^{(n+1)}[a + t(z-a)] dt$$

To simplify notation let us define R_n^a

$$R_n^a = \frac{(-1)^n (z-a)^{n+1}}{n!} \int_0^1 B_n(t) f^{(n+1)}[a + t(z-a)] dt$$

Therefore we get

$$\sum_{m=2}^n \frac{(-1)^m (z-a)^m B_m}{m!} (f^{(m)}(z) - f^{(m)}(a)) + B_0 (f(z) - f(a)) + (z-a) B_1 (f'(z) + f'(a)) = R_n^a$$

Plugging in $B_0 = 1$ and $B_1 = -\frac{1}{2}$

$$\sum_{m=2}^n \frac{(-1)^m (z-a)^m B_m}{m!} (f^{(m)}(z) - f^{(m)}(a)) + (f(z) - f(a)) - (z-a) \frac{1}{2} (f'(z) + f'(a)) = R_n^a$$

Adding $(z-a)f'(a)$ to both sides and rearranging we get

$$(z-a)f'(a) = f(z) - f(a) - \frac{(z-a)}{2} (f'(z) - f'(a)) + \sum_{m=2}^n \frac{(-1)^m (z-a)^m B_m}{m!} (f^{(m)}(z) - f^{(m)}(a)) - R_n^a$$

Define $w = z - a$ and $g(x) = f'(x)$ therefore we get

$$wg(a) = \int_{x=a}^{a+w} g(x) dx - \frac{w}{2} (g(a+w) - g(a)) + \sum_{m=2}^n \frac{(-1)^m w^m B_m}{m!} (g^{(m-1)}(a+w) - g^{(m-1)}(a)) - R_n^a$$

Allowing a to be multiples of w , $a' = a + jw$ and summing over i we get

$$\begin{aligned} \sum_{j=0}^{r-1} wg(a+jw) &= \sum_{j=0}^{r-1} \int_{x=a+jw}^{a+(j+1)w} g(x) dx - \sum_{j=0}^{r-1} \frac{w}{2} (g(a+(j+1)w) - g(a+jw)) \\ &+ \sum_{j=0}^{r-1} \sum_{m=2}^n \frac{(-1)^m w^m B_m}{m!} (g^{(m-1)}(a+(j+1)w) - g^{(m-1)}(a+jw)) - \sum_{j=0}^{r-1} R_n^{a+jw} \end{aligned}$$

Simplify and evaluating the sums we get

$$\begin{aligned} w \sum_{j=0}^{r-1} g(a+jw) &= \int_{x=a}^{a+rw} g(x) dx - \frac{w}{2} (g(a+rw) - g(a)) \\ &+ \sum_{m=2}^n \frac{(-1)^m w^m B_m}{m!} (g^{(m-1)}(a+rw) - g^{(m-1)}(a)) - \sum_{j=0}^{r-1} R_n^{a+jw} \end{aligned}$$

Rearranging

$$w \sum_{j=0}^{r-1} g(a+jw) - \int_{x=a}^{a+rw} g(x)dx = \frac{w}{2} (g(a) - g(a+rw)) \\ + \sum_{m=2}^n \frac{(-1)^m w^m B_m}{m!} (g^{(m-1)}(a+rw) - g^{(m-1)}(a)) - \sum_{j=0}^{r-1} R_n^{a+jw}$$

Using the fact that $B_{2n+1} = 0, n \geq 1$ we can simplify the sum using $m = 2k$

$$w \sum_{j=0}^{r-1} g(a+jw) - \int_{x=a}^{a+rw} g(x)dx = \frac{w}{2} (g(a) - g(a+rw)) \\ + \sum_{k=1}^{\lfloor n/2 \rfloor} \frac{w^{2k} B_{2k}}{(2k)!} (g^{(2k-1)}(a+rw) - g^{(2k-1)}(a)) - \sum_{j=0}^{r-1} R_n^{a+jw}$$

Recall the remainder term definition

$$R_n^a = \frac{(-1)^n (z-a)^{n+1}}{n!} \int_0^1 B_n(t) f^{(n+1)}[a+t(z-a)] dt$$

Plugging in the definition of $w = (z-a)$ we get

$$R_n^a = \frac{(-1)^n w^{n+1}}{n!} \int_0^1 B_n(t) f^{(n+1)}[a+tw] dt$$

Define the new remainder term

$$R_n = - \sum_{j=0}^{r-1} R_n^{a+jw}$$

Plug in the definition of R_n^{a+jw} and simplify

$$R_n = - \sum_{j=0}^{r-1} \frac{(-1)^n w^{n+1}}{n!} \int_0^1 B_n(t) f^{(n+1)}(a+jw+tw) dt$$

Simplify the remainder term using the fractional operator $\{x\}$

$$R_{n,r} = \frac{(-1)^{n+1} w^{n+1}}{n!} \int_0^r B_n(\{t\}) f^{(n+1)}(a+tw) dt$$

3.1 Infinite Series

Note that as n approaches infinity, the remainder term goes to 0.

Theorem 3.2. *If $|g^{(k)}(x)| \leq M \quad \forall x$ and a, w, M finite then*

$$\boxed{\lim_{n \rightarrow \infty} R_{n,r} = \lim_{n \rightarrow \infty} \frac{(-1)^{n+1} w^{n+1}}{n!} \int_0^r B_n(\{t\}) f^{(n+1)}(a+tw) dt = 0} \quad (5)$$

Proof

Starting from the definition of $R_{n,r}$

$$\begin{aligned}\lim_{n \rightarrow \infty} R_n &= \lim_{n \rightarrow \infty} \frac{(-1)^{n+1} w^{n+1}}{n!} \int_0^r B_n(\{t\}) f^{(n+1)}(a + tw) dt = 0 \\ \lim_{n \rightarrow \infty} R_n &\leq \lim_{n \rightarrow \infty} |R_n| \\ |R_n| &\leq \frac{|w|^{n+1}}{n!} \int_0^r |B_n(\{t\})| M dt \leq \frac{|w|^{n+1} |B_n(t)| r M}{n!}\end{aligned}$$

Using $\lim_{n \rightarrow \infty} \frac{B_n(x)}{n!} = 0$ from *BERNOULLI DOCUMENT TODO*

$$\lim_{n \rightarrow \infty} R_n \leq \lim_{n \rightarrow \infty} |R_n| = 0$$

3.2 Simple version

By specifying $a = 0$ and $w = 1$ we obtain an exact difference between the summation and the integral of any analytic function $g(x)$.

Theorem 3.3. *For any analytical $g(x)$*

$$\begin{aligned}\sum_{j=0}^r g(j) - \int_{x=0}^r g(x) dx &= \frac{g(0) + g(r)}{2} + \sum_{k=1}^{\lfloor n/2 \rfloor} \frac{B_{2k}}{(2k)!} \left(g^{(2k-1)}(r) - g^{(2k-1)}(0) \right) + R_{n,r} \\ R_{n,r} &= \frac{(-1)^{n+1}}{n!} \int_0^r B_n(\{t\}) g^{(n)}(t) dt\end{aligned}$$

(6)

Proof

Starting from the Euler-Maclaurin formula

$$\begin{aligned}w \sum_{j=0}^{r-1} g(a + jw) - \int_{x=a}^{a+rw} g(x) dx &= \frac{w}{2} (g(a) - g(a + rw)) + \\ &\sum_{k=1}^{\lfloor n/2 \rfloor} \frac{w^{2k} B_{2k}}{(2k)!} \left(g^{(2k-1)}(a + rw) - g^{(2k-1)}(a) \right) + R_{n,r} \\ R_{n,r} &= \frac{(-1)^{n+1} w^{n+1}}{n!} \int_0^r B_n(\{t\}) g^{(n)}(a + tw) dt\end{aligned}$$

Plugging in $a = 0$ and $w = 1$ we get

$$\begin{aligned}\sum_{j=0}^{r-1} g(j) - \int_{x=0}^r g(x) dx &= \frac{1}{2} (g(0) - g(r)) + \sum_{k=1}^{\lfloor n/2 \rfloor} \frac{B_{2k}}{(2k)!} \left(g^{(2k-1)}(r) - g^{(2k-1)}(0) \right) + R_{n,r} \\ R_{n,r} &= \frac{(-1)^{n+1}}{n!} \int_0^r B_n(\{t\}) g^{(n)}(t) dt\end{aligned}$$

Adding $g(r)$ to both sides

$$\sum_{j=0}^r g(j) - \int_{x=0}^r g(x) dx = \frac{g(0) + g(r)}{2} + \sum_{k=1}^{\lfloor n/2 \rfloor} \frac{B_{2k}}{(2k)!} \left(g^{(2k-1)}(r) - g^{(2k-1)}(0) \right) + R_{n,r}$$